

VANESSA THANDOLWENKOSI FAMBAREGA

BSc Computer Science Graduate

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PROFILE

Hands-on experience building data-driven applications and solving real business problems. I've worked on everything from IoT sensor networks with real-time analytics to full-stack web applications that required database optimization and performance tuning. I understand how to configure network infrastructure and I'm comfortable working across the full technology stack. I'm looking for opportunities where I can apply my technical and analytical skills to help organizations tackle complex challenges through smart, data-driven solutions.

EDUCATION

Eduvos, *Bachelor of Science in Computer Science* 02/2023 – 12/2025

- Key Modules: Object Oriented Programming | Human Computer Interaction | Advanced Network Security | Data Structures and Algorithms | Object Oriented Systems Analysis and Design | Artificial Intelligence Techniques | IT Project Management | Software Design | Web Development and e-Commerce

Awards:

- Top Academic Performer in Artificial Intelligence and Project Management.

WORK EXPERIENCE

BBD Software, *Student Vacation work* 12/2025 – 01/2026

- Full-Stack Development: Built and deployed features using React (Vite), TypeScript, and Node.js. Integrated Firebase and Supabase for real-time database management and authentication.
- Cloud Integration: Worked with Google Cloud Platform to handle backend services and application hosting.
- Team Participation: Participated in daily stand-ups and team sprints. Gained experience in professional communication and keeping stakeholders updated on project progress.
- Technical Upskilling: Learned and implemented Docker for containerization and used advanced Git workflows to manage code versions within a large team.
- Problem Solving: Worked through pair-programming sessions to find and fix bugs in existing code and improve overall app performance.
- Workflow Management: Completed technical tasks under strict production deadlines, moving from academic theory to real-world software delivery.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Coding Club — Active member | 2024-2025 | **Golden Key International Honor Society** — |

Invited member 2025

TECHNICAL SKILLS

Programming — C#, Java (OOP), PHP, JavaScript, TypeScript, HTML/CSS, C++, SQL,
Frameworks/Tools — React (Vite), Node.js, Express, Docker, Git/GitHub, Harness.io., **Databases & Cloud** — MySQL, PostgreSQL, MongoDB, Firebase, Supabase, Google Cloud Platform.,
Hardware/IoT — Arduino, Raspberry Pi., **Networking** — Cisco Packet Tracer, TCP/IP, Routing & Switching.

PROJECTS

Rand Merchant Bank | GirlCode Hackathon 18/03/2026 – 19/03/2026

- Lending Automation MVP: Designed a digital solution for RMB to automate document-heavy corporate lending workflows, replacing manual "Capture Me" tasks to reduce operational risk.
- Full-Stack Build: Developed the core prototype using Python (Flask), PostgreSQL, and JavaScript, featuring a dashboard to track 20-year compliance milestones and automated task generation.
- Safety of the Bank: Integrated Harness.io to implement secure CI/CD pipelines and automated security gates, ensuring financial data integrity and compliant deployment practices.

Learning Management System (Capstone Project) 11/2025

- Developing scalable LMS for matric rewrite students with comprehensive analytics and business intelligence features including student progress tracking system with real-time performance dashboards and data visualizations.
- Implemented analytics engine to identify learning trends, completion rates, at-risk students, e-commerce functionality with sales reporting and revenue tracking for course enrollment
- Designed database-driven architecture supporting multi-user access with query optimization for 50+ concurrent users.

Small Streets e-Commerce Platform 06/2025

- Built C2C marketplace with secure user authentication, product listings, shopping cart, and order processing
- Designed for South African market with localized payment solutions based on market research and consumer data analysis
- Used data visualization tools to monitor platform performance metrics and identify growth opportunities

IoT Smart Home System 08/2024

- Designed automated home prototype with comprehensive sensor data collection and analytics
- Deployed motion and temperature sensors with real-time monitoring and data visualization dashboards
- Analyzed sensor data trends to optimize automated lighting control and reduce energy consumption
- Integrated hardware-software systems from concept to working prototype with data logging for performance analysis

LANGUAGES

English | Zulu